

Data Collection and Preprocessing Phase

Date	15 March 2024
Team ID	XXXXXX
Project Title	XXXXXX
Maximum Marks	10 Marks

Data Exploration and Preprocessing Template

Identifies data sources, assesses quality issues like missing values and duplicates, and implements resolution plans to ensure accurate and reliable analysis.

Section	Description
Data Overview	The dataset contains 100 records capturing plant growth stages, environmental factors (temperature, humidity, soil moisture), and management activities (fertilizer used, water given). The goal is to predict growth stages based on these features.
Data Cleaning	Checked and handled missing values by applying mean/mode imputation where necessary. Verified and removed duplicate records. Corrected minor inconsistencies in text fields (e.g., Growth Stage labels).
Data Transformation	Used Power Query for filtering irrelevant records, sorting data by Date and Growth Stage, pivoting tables for analysis, and creating calculated columns like "Management Effort" (sum of water and fertilizer inputs).
Data Type Conversion	Converted Date fields to proper Date/Time format, numeric fields (Temperature, Humidity, Soil Moisture) to Decimal format, and Growth Stage fields to Text format.

Column Splitting and Merging	Split Date fields into Month and Year for seasonal analysis. Merged Water and Fertilizer columns to create a "Total Management Effort" metric for better insights.
Data Modeling	Defined logical relationships among fields (e.g., Plant ID to Growth Stage progression). Created basic DAX measures for Average Temperature, Average Humidity, and Total Water Usage.
Save Processed Data	Saved the cleaned and transformed dataset into a new Excel file, ensuring it is ready for visualization and analysis in Power BI.